Digital Logic Gates

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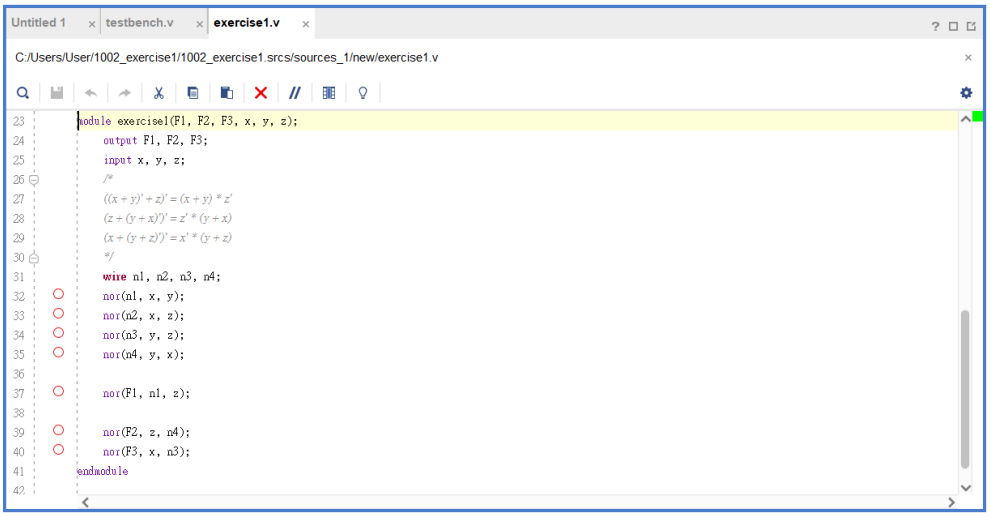
實驗內容

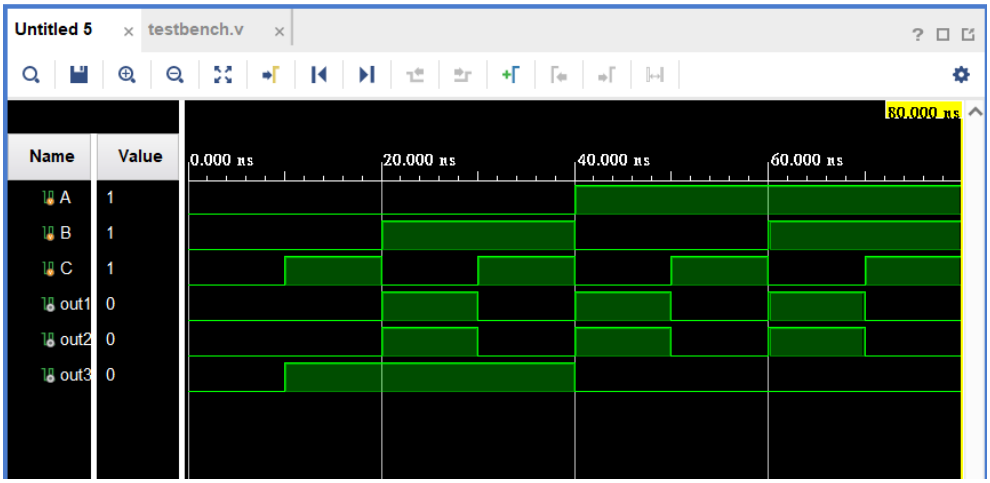
Exercise1

Derive (x ↓ y) ↓ z = (x + y)z', z ↓ (y ↓ x) = z' (y + x) and

x ↓ (y ↓ z) = x' (y + z) (applying DeMorgan's theorem)

1. Verify (x ↓ y) ↓ z = z ↓ (y ↓ x) (using **Structural level modeling**)
2. Verify (x ↓ y) ↓ z ≠ x ↓ (y ↓ z) (using **Structural level modeling**)





跟據題目， $(x \downarrow y) \downarrow z = \mathbf{(x + y)z'}$

$z \downarrow (y \downarrow x) = \mathbf{z' (y + x)}$

$x \downarrow (y \downarrow z) = \mathbf{x' (y + z)}$。

第一步要驗證 $(x + y)z'$ 是否等於 $z' (y + x)$ ，由實驗結果的波形圖可以證明等式 $(x + y)z' = z' (y + x)$ 成立。

我們也可以從布林代數的定理進行二次驗證：

Postulate 4a, $(x + y)z' = xz' + yz'$, $z' (y + x) = z'y + z'x$

Postulate 3b, $z'y + z'x = yz' + xz'$

Postulate 3a, $yz' + xz' = xz' + yz'$

$\Longrightarrow (x + y)z' = xz' + yz' = xz' + yz' = z' (y + x)$

第二步則是要驗證 $(x + y)z'$ 是否不等於 $x' (y + z)$，實驗結果的波形圖可以發現該等式式不成立的。

我們也試著從布林代數的定理進行推算看看：

Postulate 4a, $(x + y)z' = xz' + yz'$, $x' (y + z) = x'y + x'z$

Postulate 3a, $x'y + x'z = x'z + x'y$

$\Longrightarrow (x + y)z' = xz' + yz' \neq x'z + x'y = x' (y + z)$

推導 $(x \downarrow y) \downarrow z = (x + y)z'$ $\downarrow$: nor gate

$x \downarrow y = (x + y)' = x' y'$, by Thm 5a (DeMorgan Thm)

$\Longrightarrow (x \downarrow y) \downarrow z = [(x + y)' + z]'$

$= [x' y' + z]' = (x' y')' z'$

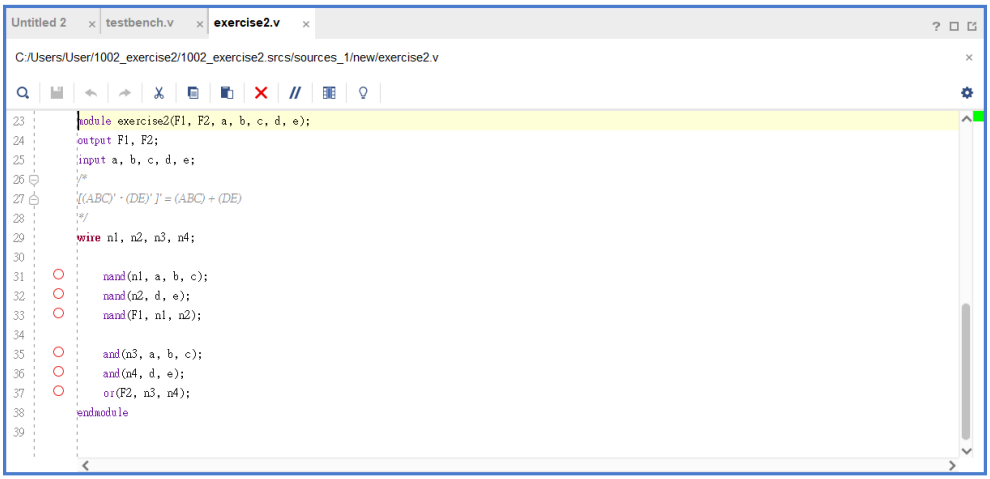
$= (x + y) z'$, by Thm 5b (DeMorgan Thm)

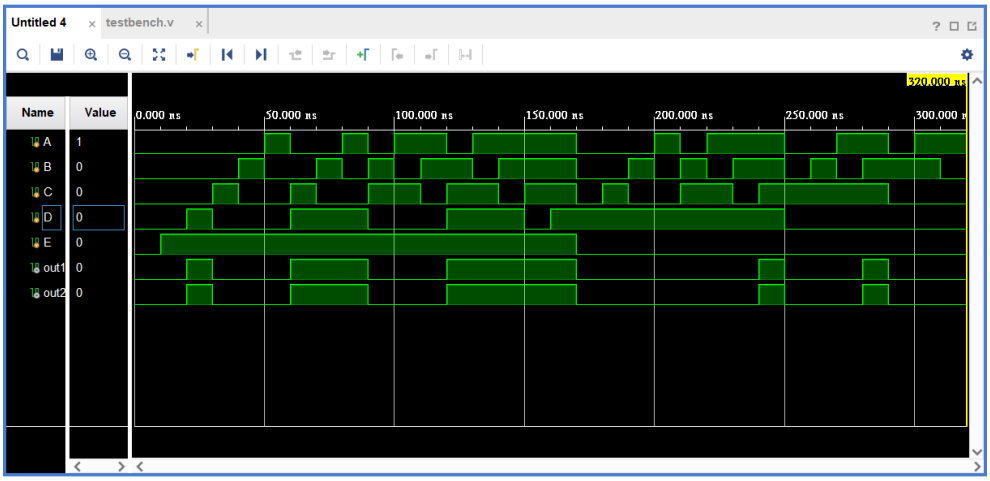
$\therefore$ **$(x \downarrow y) \downarrow z = (x + y) z'$**

Exercise2

Derive $[(ABC)' \times (DE)']' = ABC + DE$ (applying DeMorgan's theorem)

Use **Structural level modeling** to verify that $[(ABC)' \times (DE)']' = ABC + DE$





由實驗結果可以驗證 $[(ABC)' \times (DE)']' = ABC + DE$。

以布林代數定理驗證過程如下：

(ABC)' = A' + B' + C', (DE)' = D' + E', by DeMorgan's theorem

[(ABC)' × (DE)']' = [(A' + B' + C')(D' + E')]'

= (A' + B' + C')' + (D' + E')'

= **ABC + DE**, by DeMorgan's theorem

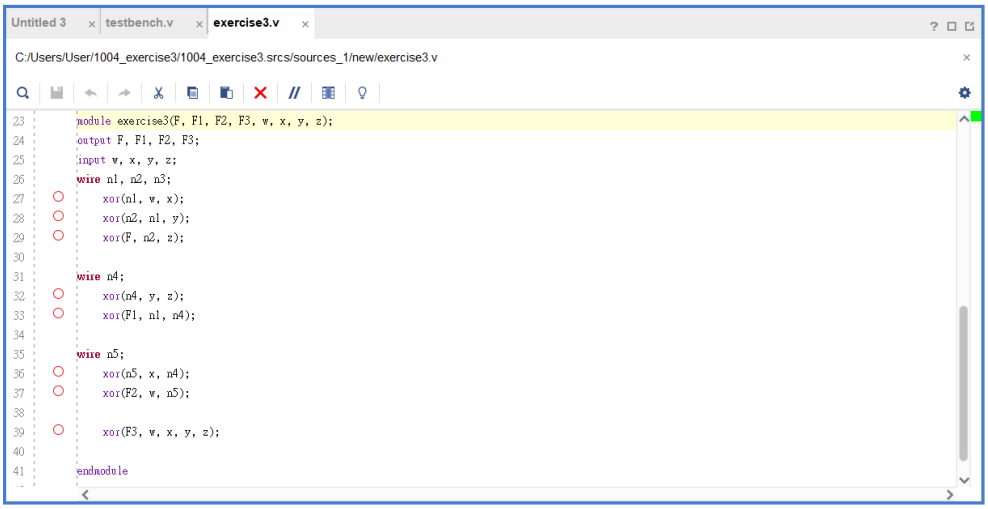
Exercise3

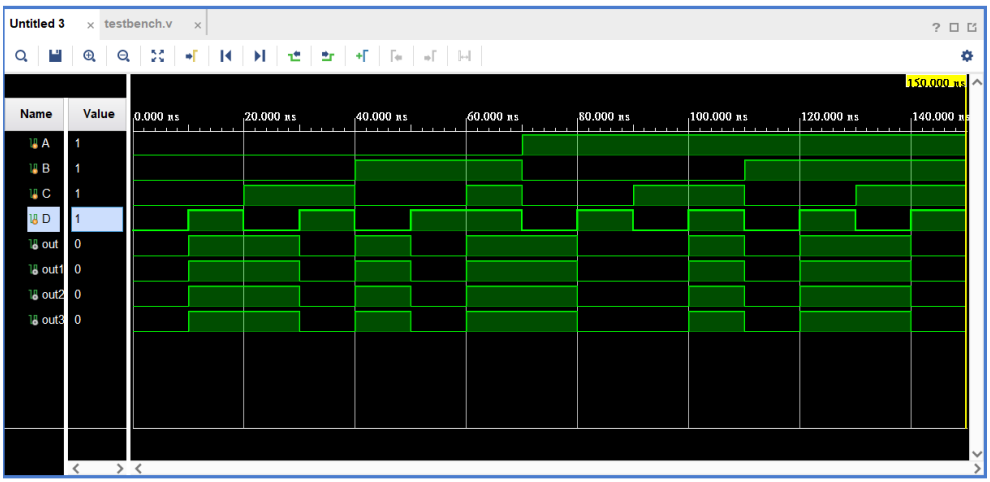
XOR is an odd function.

Complete the truth table of four-variable XOR function:

1. F(w, x, y, z)
2. F1 = (w ⊕ x) ⊕ (y ⊕ z)
3. F2 = w ⊕ (x ⊕ (y ⊕ z))
4. F3 = w ⊕ x ⊕ y ⊕ z

Verify F1 = F2 = F3 (using **Structural level modeling**)





由實驗結果可以驗證 F1 = F2 = F3。

使用布林代數定理驗證過程如下：

XOR gate: $w \oplus x = wx' + w'x$

$\Longrightarrow$ F1 $= \mathbf{(w \oplus x) \oplus (y \oplus z)} \qquad = (wx' + w'x) \oplus (yz' + y'z)$

$= (wx' + w'x)\,(yz' + y'z)' + (wx' + w'x)'\,(yz' + y'z)$

$= (wx' + w'x)\,(yz')'(y'z)' + (wx')'(w'x)'\,(yz' + y'z)$

$= (wx' + w'x)\,(yz + y'z') + (wx + w'x')\,(yz' + y'z)$

$= \mathbf{wx'yz + w'xyz + wx'y'z' + w'xy'z' + wxyz' + w'x'yz' + wxy'z + w'x'y'z}$

$\Longrightarrow$ F2 $= \mathbf{w \oplus (x \oplus (y \oplus z))} \qquad = w \oplus (x \oplus (yz' + y'z))$

$= w \oplus (\,x'(yz' + y'z) + x(yz' + y'z)'\,)$

$= w \oplus (\,x'yz' + x'y'z + x(yz')'(y'z)'\,)$

$= w \oplus (\,x'yz' + x'y'z + x(y' + z)(y + z')\,)$

$= w \oplus (\,x'yz' + x'y'z + x(y'y + yz + y'z' + zz')$

$= w \oplus (x'yz' + x'y'z + xyz + xy'z')$

$= w'(x'yz' + x'y'z + xyz + xy'z') + w(x'yz' + x'y'z + xyz + xy'z')'$ = ...*(用筆算，超麻煩)*.....

$=$ **$w'x'yz' + w'x'y'z + w'xyz + w'xy'z' + wx'yz + wxy'z + wxyz' + wx'y'z'$**

$\Longrightarrow F3 = w \oplus x \oplus y \oplus z = (w'x + wx') \oplus y \oplus z$

$= [(w'x + wx')'y + (w'x + wx')y'] \oplus z$

$= (w'x'y + wxy + w'x y' + wx'y') \oplus z$

$= (w'x'y + wxy + w'x y' + wx'y')'z + (w'x'y + wxy + w'x y' + wx'y')z'$

$= (w + x + y') (w' + x' + y') (w + x' + y) (w' + x + y)z + (w'x'y + wxy + w'x y' + wx'y')z'$

$= (w'x + wx' + y')(w'x' + wx + y)z + w'x'yz' + wxyz' + w'xy'z' + wx'y'z'$

$=$ **$w'x'y'z + wxy'z + w'xyz + wx'yz + w'x'yz' + wxyz' + w'xy'z' + wx'y'z'$**

$\therefore$ **F1 = F2 = F3** $= w'x'yz' + w'x'y'z + w'xyz + w'xy'z' + wx'yz + wxy'z + wxyz' + wx'y'z'$

實驗心得

這次實驗總體而言進行得比上次順利很多，也是我從開學以來第一次提早完成，小有成就感。

此次實驗首次使用到邏輯閘：XOR，原本以為這個特殊的 gate 會比較難使用，但實驗時發現撰寫 verilog 的 XOR gate 並沒有想像中的那麼困難；反倒是在寫實驗心得時，用布林代數去驗證 XOR gate 的等式非常的麻煩，計算過程一大堆。而且此次實驗才使用到 4 個 XOR 閘而已，如果以後用到更多個時，我無法想像那驗證過程的計算量.....。